

Integration by Substitution

Date_____ Period_____

Evaluate each indefinite integral. Use the provided substitution.

1) $\int \frac{20x^3}{\sqrt{25 - 25x^8}} dx; \quad u = 5x^4$

2) $\int \frac{10x^4}{9 + 4x^{10}} dx; \quad u = 2x^5$

3) $\int -\frac{2 \cdot \csc^2 2x}{\cot(2x) \cdot \sqrt{\cot^2 2x - 1}} dx; \quad u = \cot 2x$

4) $\int \frac{1}{x\sqrt{25 - (\ln -2x)^2}} dx; \quad u = \ln -2x$

Evaluate each indefinite integral.

5) $\int \frac{8x}{\sqrt{9 - 16x^4}} dx$

6) $\int \frac{3x^2}{x^3 \sqrt{x^6 - 1}} dx$

7) $\int \frac{10x}{16 + 25x^4} dx$

8) $\int -\frac{4\sin 4x}{\sqrt{9 - \cos^2 4x}} dx$